SMART INDIA HACKATHON (SIH) 2026 • MINISTRY OF RAILWAYS, GOVT. OF INDIA

AI-Powered Automatic Block Planning System

Problem Statement 26027: To Maximize Asset Availability for Train Operations on Indian Railways

Problem Statement ID:	26027 (Ministry of Railways / AICTE)
Category / Domain:	Smart Transportation, AI & Operations Research, Real-Time Digital Twin
Core Architecture:	TMS/SMMS/TDMS → AI Criticality → DBSCAN Clustering → OR-Tools CP-SAT → FastAPI → React 19
Target Beneficiaries:	Section Controllers, Divisional Operations & Engineering Maintenance Wings

1. Introduction

Indian Railways operates one of the world's largest and most congested rail networks, running over 21,000 passenger and freight trains daily across 68,000 route kilometers. To maintain uncompromising safety standards and prevent catastrophic derailments, regular maintenance of tracks, signaling systems, and overhead electrical (OHE) lines is mandatory. However, maintenance requires "traffic and power blocks" during which train movements on specific tracks must be halted or diverted. Currently, block planning across Indian Railways divisions is largely managed through manual, negotiation-heavy meetings. Without automated coordination against live train timetables, maintenance requests conflict with high-priority traffic, causing severe train delays, underutilized block windows, and repeated corridor shutdowns.

2. Problem Statement

In Indian Railways, maintenance demands from the Track (Civil/TMS), Signalling (S&T/SMMS), and Electrical (TRD/TDMS) departments reside in isolated, independent systems. Because these departments schedule work separately without real-time alignment with the Control Office Application (COA) train timetables, the same track section is often shut down repeatedly on consecutive days by different teams. This lack of coordination throttles line capacity, causes cascading delays to high-priority trains like Vande Bharat and Rajdhani, and prevents rapid dynamic rescheduling when unexpected rail defects or severe weather events strike in real time.

3. Aim of the Project

The primary aim of this project is to develop an intelligent, automated block scheduling and real-time digital twin platform for Indian Railways. The platform coordinates multi-departmental maintenance demands, maximizes track availability for commercial train operations, eliminates scheduling conflicts, and dynamically reschedules maintenance windows within seconds during real-time track disruptions or adverse weather conditions.

4. Project Description

This project provides an end-to-end decision-support platform for railway section controllers by combining machine learning, spatial clustering, and mathematical constraint programming. The system ingests defect records across Track, Signal, and Electrical departments and ranks their urgency using an AI Criticality Scoring engine (XGBoost blended with safety heuristics) based on defect severity, speed restrictions, and passenger traffic density. It then applies spatial clustering (DBSCAN) to group overlapping tasks along the same corridor (within 2.0 km), allowing multiple departments to execute repairs simultaneously under a single integrated block.

The scheduling core employs Google OR-Tools CP-SAT to calculate mathematically conflict-free weekly block schedules that respect all train headways, crew limits, and maintenance machinery availability. Furthermore, the platform incorporates a Real-Time Railway Digital Twin that ingests live weather conditions via the Open-Meteo API and streams simulated IoT track vibration telemetry. When an anomaly or emergency defect is detected, the system calculates the cascading blast radius and generates an updated schedule within 60 seconds. The entire platform is accessible through an interactive React 19 Web GIS dashboard featuring live track maps, Gantt timelines, and one-click PDF Block Sanction Order generation with QR verification.

5. Key Objectives

- **Automated Conflict-Free Planning:** Replace days of manual coordination by automatically computing conflict-free corridor block schedules in under 2 minutes.
- **Maximize Track Availability:** Elevate active track availability for train operations on high-density corridors to over 88.5% while adhering to all safety protocols.
- **Multi-Department Integration:** Achieve a cross-departmental integration rate of over 70%, bundling track, signal, and electrical maintenance into joint windows.
- **Real-Time Anomaly Resilience:** Detect rail vibration and thermal anomalies and dynamically generate conflict-free replacement schedules in under 60 seconds.
- **Controller Decision Support:** Equip Section Controllers with an interactive GIS map, schedule timeline, and instant export of official block sanction permits.

6. References

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